JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Complex Numbers

JEE Main Mathematics · Previous Year Questions with Answer Key

205	177	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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MCQ (Single Correct Answer)	167 questions
Numerical	38 questions
Answer key	at the end

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.

MCQ (Single Correct Answer) 167 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

The number of values of $z \in \mathbb{C}$, satisfying the equations $|z-(4+8i)|=\sqrt{10}$ and $|z-(3+5i)|+|z-(5+11i)|=4\sqrt{5}$, is :

(A) 0 (B) 2 (C) 1 (D) 4

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Let $S=\left\{z \in \mathbb{C}: z^2+\sqrt{6}iz-3=0\right\}$. Then $\sum\limits_{z \in S} z^8$ is equal to :

(A) 162 (B) 184 (C) 262 (D) 324

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z}+2+i)+k(2+3i)=0, z \in \mathrm{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha+\beta)$ is equal to:

(A) -10 (B) -8 (C) $10\sqrt{13}$ (D) $8\sqrt{13}$

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Let $z_1, z_2 \in \mathbb{C}$ be the distinct solutions of the equation $z^2+4z-(1+12i)=0$.

Then $|z_1|^2+|z_2|^2$ is equal to :

(A) 18 (B) 22 (C) 29 (D) 34

Q5. JEE Main 2026 (Online) 5th April Morning Shift

Let $a, b \in \mathbb{C}$. Let α, β be the roots of the equation $x^2+ax+b=0$. If $\beta-\alpha=\sqrt{11}$ and $\beta^2-\alpha^2=3i\sqrt{11}$, then $\left(\beta^3-\alpha^3\right)^2$ is equal to:

(A) 160 (B) 176 (C) 194 (D) 187

Q6. JEE Main 2026 (Online) 4th April Evening Shift

Let $\mathrm{S}=\left\{z \in \mathrm{C}: z^2+4z+16=0\right\}$. Then $\sum\limits_{z \in \mathrm{S}}|z+\sqrt{3}\mathrm{i}|^2$ is equal to :

(A) 42 (B) 23 (C) 27 (D) 38

Q7. JEE Main 2026 (Online) 4th April Morning Shift

Let z be a complex number such that $|z+2|=|z-2|$ and $\arg\left(\frac{z+3}{z-i}\right)=\frac{\pi}{4}$. Then $|z|^2$ is equal to:

(A) 9 (B) 4 (C) 5 (D) 1

Q8. JEE Main 2026 (Online) 2nd April Evening Shift

Let the circles $C_1 : |z| = r$ and $C_2 : |z - 3 - 4i| = 5$, $z \in \mathbb{C}$, be such that C_2 lies within C_1.
If z_1 moves on C_1, z_2 moves on C_2 and $\min |z_1 - z_2| = 2$, then $\max |z_1 - z_2|$ is equal to :

(A) 12 (B) 17 (C) 22 (D) 24

Q9. JEE Main 2026 (Online) 2nd April Morning Shift

Let x and y be real numbers such that $50\left(\frac{2x}{1+3i} - \frac{y}{1-2i}\right) = 31 + 17i, i = \sqrt{-1}$. Then the value of $10(x - 3y)$ is :

(A) 20 (B) 31 (C) 35 (D) 75

Q10. JEE Main 2026 (Online) 28th January Evening Shift

Let

$A = \{z \in \mathbb{C} : |z - 2| \leq 4\}$ and

$B = \{z \in \mathbb{C} : |z - 2| + |z + 2| = 5\}$.

Then the max $\{|z_1 - z_2| : z_1 \in A$ and $z_2 \in B\}$ is :

(A) $\frac{17}{2}$ (B) 8 (C) 9 (D) $\frac{15}{2}$

Q11. JEE Main 2026 (Online) 28th January Morning Shift

Let z be a complex number such that $|z - 6| = 5$ and $|z + 2 - 6i| = 5$. Then the value of $z^3 + 3z^2 - 15z + 141$ is equal to :

(A) 61 (B) 37 (C) 42 (D) 50

Q12. JEE Main 2026 (Online) 24th January Morning Shift

Let $S = \left\{z \in \mathbb{C} : \left|\frac{z-6i}{z-2i}\right| = 1 \text{ and } \left|\frac{z-8+2i}{z+2i}\right| = \frac{3}{5}\right\}$.

Then $\sum\limits_{z \in S} |z|^2$ is equal to :

(A) 413 (B) 398 (C) 385 (D) 423

Q13. JEE Main 2026 (Online) 23rd January Evening Shift

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}, i = \sqrt{-1}$, then $\left(z^{201} - i\right)^8$ is equal to

(A) 1 (B) 0 (C) -1 (D) 256

Q14. JEE Main 2026 (Online) 23rd January Morning Shift

Let $S = \{z : 3 \leqslant |2z - 3(1 + i)| \leqslant 7\}$ be a set of complex numbers.

Then $\text{Min}_{z \in S} \left|\left(z + \frac{1}{2}(5 + 3i)\right)\right|$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{5}{2}$ (C) 2 (D) $\frac{3}{2}$

Q15. JEE Main 2026 (Online) 22nd January Evening Shift

Let $S = \{z \in \mathbb{C} : 4z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} |z|^2$ is equal to:

(A) $\frac{5}{64}$ (B) $\frac{1}{16}$ (C) $\frac{7}{64}$ (D) $\frac{3}{16}$

Q16. JEE Main 2026 (Online) 21st January Evening Shift

Let z be the complex number satisfying $|z - 5| \leq 3$ and having maximum positive principal argument.

Then $34\left|\frac{5z-12}{5iz+16}\right|^2$ is equal to:

(A) 20 (B) 26 (C) 12 (D) 16

Q17. JEE Main 2026 (Online) 21st January Morning Shift

If $x^2 + x + 1 = 0$, then the value of $\left(x + \frac{1}{x}\right)^4 + \left(x^2 + \frac{1}{x^2}\right)^4 + \left(x^3 + \frac{1}{x^3}\right)^4 + \ldots + \left(x^{25} + \frac{1}{x^{25}}\right)^4$ is:

(A) 162 (B) 145 (C) 128 (D) 175

Q18. JEE Main 2025 (Online) 8th April Evening Shift

Let $A = \left\{\theta \in [0, 2\pi] : 1 + 10\,\mathrm{Re}\left(\frac{2\cos\theta + i\sin\theta}{\cos\theta - 3i\sin\theta}\right) = 0\right\}$. Then $\sum_{\theta \in A} \theta^2$ is equal to

(A) $\frac{21}{4}\pi^2$ (B) $6\pi^2$ (C) $\frac{27}{4}\pi^2$ (D) $8\pi^2$

Q19. JEE Main 2025 (Online) 7th April Evening Shift

If the locus of z $\in \mathbb{C}$, such that $\mathrm{Re}\left(\frac{z-1}{2z+i}\right) + \mathrm{Re}\left(\frac{\bar{z}-1}{2\bar{z}-i}\right) = 2$, is a circle of radius r and center (a, b), then $\frac{15ab}{r^2}$ is equal to :

(A) 16 (B) 24 (C) 12 (D) 18

Q20. JEE Main 2025 (Online) 7th April Morning Shift

Among the statements

(S1) : The set $\{z \in \mathbb{C} - \{-i\} : |z| = 1$ and $\frac{z-i}{z+i}$ is purely real $\}$ contains exactly two elements, and

(S2) : The set $\{z \in \mathbb{C} - \{-1\} : |z| = 1$ and $\frac{z-1}{z+1}$ is purely imaginary $\}$ contains infinitely many elements.

(A) both are incorrect (B) both are correct

(C) only (S2) is correct (D) only (S1) is correct

Q21. JEE Main 2025 (Online) 4th April Evening Shift

Let the product of $\omega_1 = (8+i)\sin\theta + (7+4i)\cos\theta$ and $\omega_2 = (1+8i)\sin\theta + (4+7i)\cos\theta$ be $\alpha + i\beta$, $i = \sqrt{-1}$. Let p and q be the maximum and the minimum values of $\alpha + \beta$ respectively. Then $\mathrm{p} + \mathrm{q}$ is equal to :

(A) 130 (B) 150 (C) 160 (D) 140

Q22. JEE Main 2025 (Online) 3rd April Evening Shift

If $z_1, z_2, z_3 \in \mathbb{C}$ are the vertices of an equilateral triangle whose centroid is z_0, then $\sum_{k=1}^{3} (z_k - z_0)^2$ is equal to

(A) 0 (B) 1 (C) i (D) -i

Q23. JEE Main 2025 (Online) 3rd April Morning Shift

Let $z \in C$ be such that $\frac{z^2+3i}{z-2+i} = 2+3i$. Then the sum of all possible values of z^2 is :

(A) $-19+2i$ (B) $-19-2i$ (C) $19-2i$ (D) $19+2i$

Q24. JEE Main 2025 (Online) 2nd April Morning Shift

Let z be a complex number such that $|z| = 1$. If $\frac{2+\mathrm{k}^2 z}{\mathrm{k}+\bar{z}} = \mathrm{k}z, \mathrm{k} \in \mathbf{R}$, then the maximum distance of $\mathrm{k} + i\mathrm{k}^2$ from the circle $|z-(1+2i)| = 1$ is :

(A) $\sqrt{5}+1$ (B) 3 (C) $\sqrt{3}+1$ (D) 2

Q25. JEE Main 2025 (Online) 29th January Morning Shift

Let $|z_1 - 8 - 2i| \leq 1$ and $|z_2 - 2 + 6i| \leq 2$, $z_1, z_2 \in \mathbb{C}$. Then the minimum value of $|z_1 - z_2|$ is :

(A) 3 (B) 10 (C) 7 (D) 13

Q26. JEE Main 2025 (Online) 28th January Evening Shift

If $\alpha + i\beta$ and $\gamma + i\delta$ are the roots of $x^2 - (3-2i)x - (2i-2) = 0$, $i = \sqrt{-1}$, then $\alpha\gamma + \beta\delta$ is equal to:

(A) 2 (B) -6 (C) 6 (D) -2

Q27. JEE Main 2025 (Online) 28th January Morning Shift

Let O be the origin, the point A be $z_1 = \sqrt{3} + 2\sqrt{2}i$, the point $B(z_2)$ be such that $\sqrt{3}|z_2| = |z_1|$ and $\arg(z_2) = \arg(z_1) + \frac{\pi}{6}$. Then

(A) area of triangle ABO is $\frac{11}{4}$

(B) area of triangle ABO is $\frac{11}{\sqrt{3}}$

(C) ABO is a scalene triangle

(D) ABO is an obtuse angled isosceles triangle

Q28. JEE Main 2025 (Online) 24th January Morning Shift

If α and β are the roots of the equation $2z^2 - 3z - 2i = 0$, where $i = \sqrt{-1}$, then $16 \cdot \mathrm{Re}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right) \cdot \mathrm{Im}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right)$ is equal to

(A) 441 (B) 312 (C) 409 (D) 398

Q29. JEE Main 2025 (Online) 23rd January Evening Shift

The number of complex numbers z, satisfying $|z| = 1$ and $\left|\frac{z}{\bar{z}} + \frac{\bar{z}}{z}\right| = 1$, is :

(A) 8 (B) 10 (C) 4 (D) 6

Q30. JEE Main 2025 (Online) 23rd January Morning Shift

Let $\left|\frac{\bar{z}-i}{2\bar{z}+i}\right| = \frac{1}{3}, z \in C$, be the equation of a circle with center at C. If the area of the triangle, whose vertices are at the points $(0, 0), C$ and $(\alpha, 0)$ is 11 square units, then α^2 equals:

(A) $\frac{121}{25}$ (B) 100 (C) $\frac{81}{25}$ (D) 50

Q31. JEE Main 2025 (Online) 22nd January Evening Shift

Let the curve $z(1+i) + \bar{z}(1-i) = 4, z \in C$, divide the region $|z-3| \leq 1$ into two parts of areas α and β. Then $|\alpha - \beta|$ equals :

(A) $1 + \frac{\pi}{3}$ (B) $1 + \frac{\pi}{6}$ (C) $1 + \frac{\pi}{2}$ (D) $1 + \frac{\pi}{4}$

Q32. JEE Main 2025 (Online) 22nd January Morning Shift

Let z_1, z_2 and z_3 be three complex numbers on the circle $|z| = 1$ with $\arg(z_1) = \frac{-\pi}{4}$, $\arg(z_2) = 0$ and $\arg(z_3) = \frac{\pi}{4}$. If $|z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1|^2 = \alpha + \beta\sqrt{2}, \alpha, \beta \in Z$, then the value of $\alpha^2 + \beta^2$ is :

(A) 41 (B) 29 (C) 24 (D) 31

Q33. JEE Main 2024 (Online) 9th April Evening Shift

Let z be a complex number such that the real part of $\frac{z-2i}{z+2i}$ is zero. Then, the maximum value of $|z - (6+8i)|$ is equal to

(A) 8 (B) 12 (C) 10 (D) ∞

Q34. JEE Main 2024 (Online) 8th April Evening Shift

The sum of all possible values of $\theta \in [-\pi, 2\pi]$, for which $\frac{1+i\cos\theta}{1-2i\cos\theta}$ is purely imaginary, is equal to :

(A) 4π (B) 3π (C) 2π (D) 5π

Q35. JEE Main 2024 (Online) 8th April Morning Shift

Let z be a complex number such that $|z+2|=1$ and $\operatorname{Im}\left(\frac{z+1}{z+2}\right)=\frac{1}{5}$. Then the value of $|\operatorname{Re}(\overline{z+2})|$ is

(A) $\frac{2\sqrt{6}}{5}$ (B) $\frac{24}{5}$ (C) $\frac{\sqrt{6}}{5}$ (D) $\frac{1+\sqrt{6}}{5}$

Q36. JEE Main 2024 (Online) 8th April Morning Shift

If the set $R=\{(a,b): a+5b=42, a,b\in\mathbb{N}\}$ has m elements and $\sum_{n=1}^{m}\left(1-i^{n!}\right)=x+iy$, where $i=\sqrt{-1}$, then the value of $m+x+y$ is

(A) 12 (B) 4 (C) 8 (D) 5

Q37. JEE Main 2024 (Online) 6th April Evening Shift

If z_1, z_2 are two distinct complex numbers such that $\left|\frac{z_1-2z_2}{\frac{1}{2}-z_1\bar{z}_2}\right|=2$, then

(A) either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1.

(B) z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1.

(C) either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$.

(D) both z_1 and z_2 lie on the same circle.

Q38. JEE Main 2024 (Online) 5th April Evening Shift

Let $S_1=\{z\in\mathbf{C}:|z|\le 5\}, S_2=\left\{z\in\mathbf{C}:\operatorname{Im}\left(\frac{z+1-\sqrt{3}i}{1-\sqrt{3}i}\right)\ge 0\right\}$ and $S_3=\{z\in\mathbf{C}:\operatorname{Re}(z)\ge 0\}$. Then the area of the region $S_1\cap S_2\cap S_3$ is :

(A) $\frac{125\pi}{24}$ (B) $\frac{125\pi}{6}$ (C) $\frac{125\pi}{12}$ (D) $\frac{125\pi}{4}$

Q39. JEE Main 2024 (Online) 5th April Morning Shift

Consider the following two statements :

Statement I: For any two non-zero complex numbers $z_1, z_2, (|z_1|+|z_2|)\left|\frac{z_1}{|z_1|}+\frac{z_2}{|z_2|}\right|\le 2(|z_1|+|z_2|)$, and

Statement II : If x, y, z are three distinct complex numbers and $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are three positive real numbers such that $\frac{\mathrm{a}}{|y-z|}=\frac{\mathrm{b}}{|z-x|}=\frac{\mathrm{c}}{|x-y|}$, then $\frac{\mathrm{a}^2}{y-z}+\frac{\mathrm{b}^2}{z-x}+\frac{\mathrm{c}^2}{x-y}=1$.

Between the above two statements,

(A) both Statement I and Statement II are incorrect.

(B) Statement I is correct but Statement II is incorrect.

(C) Statement I is incorrect but Statement II is correct.

(D) both Statement I and Statement II are correct.

Q40. JEE Main 2024 (Online) 4th April Evening Shift

The area (in sq. units) of the region $S = \{z \in \mathbb{C} : |z-1| \leq 2; (z+\bar{z}) + i(z-\bar{z}) \leq 2, \text{Im}(z) \geq 0\}$ is

(A) $\frac{7\pi}{4}$ (B) $\frac{3\pi}{2}$ (C) $\frac{7\pi}{3}$ (D) $\frac{17\pi}{8}$

Q41. JEE Main 2024 (Online) 4th April Morning Shift

Let α and β be the sum and the product of all the non-zero solutions of the equation $(\bar{z})^2 + |z| = 0, z \in C$. Then $4(\alpha^2 + \beta^2)$ is equal to :

(A) 4 (B) 2 (C) 6 (D) 8

Q42. JEE Main 2024 (Online) 1st February Evening Shift

If z is a complex number such that $|z| \leqslant 1$, then the minimum value of $\left|z + \frac{1}{2}(3+4i)\right|$ is :

(A) 2 (B) $\frac{5}{2}$ (C) $\frac{3}{2}$ (D) 3

Q43. JEE Main 2024 (Online) 1st February Morning Shift

Let $\mathrm{S} = \{\mathrm{z} \in \mathrm{C} : |z-1| = 1$ and $(\sqrt{2}-1)(z+\bar{z}) - i(z-\bar{z}) = 2\sqrt{2}\}$. Let $z_1, z_2 \in \mathrm{S}$ be such that $|z_1| = \max_{z \in S} |z|$ and $|z_2| = \min_{z \in S} |z|$. Then $\left|\sqrt{2}z_1 - z_2\right|^2$ equals :

(A) 1 (B) 4 (C) 3 (D) 2

Q44. JEE Main 2024 (Online) 31st January Evening Shift

Let z_1 and z_2 be two complex numbers such that $z_1 + z_2 = 5$ and $z_1^3 + z_2^3 = 20 + 15i$ Then, $\left|z_1^4 + z_2^4\right|$ equals -

(A) $15\sqrt{15}$ (B) $30\sqrt{3}$ (C) $25\sqrt{3}$ (D) 75

Q45. JEE Main 2024 (Online) 30th January Evening Shift

If z is a complex number, then the number of common roots of the equations $z^{1985} + z^{100} + 1 = 0$ and $z^3 + 2z^2 + 2z + 1 = 0$, is equal to

(A) 0 (B) 2 (C) 1 (D) 3

Q46. JEE Main 2024 (Online) 30th January Morning Shift

If $z = x + iy, xy \neq 0$, satisfies the equation $z^2 + i\bar{z} = 0$, then $\left|z^2\right|$ is equal to :

(A) 9 (B) $\frac{1}{4}$ (C) 4 (D) 1

Q47. JEE Main 2024 (Online) 29th January Evening Shift

Let r and θ respectively be the modulus and amplitude of the complex number $z = 2 - i\left(2\tan\frac{5\pi}{8}\right)$, then (r, θ) is equal to

(A) $\left(2\sec\frac{11\pi}{8}, \frac{11\pi}{8}\right)$ (B) $\left(2\sec\frac{3\pi}{8}, \frac{3\pi}{8}\right)$

(C) $\left(2\sec\frac{5\pi}{8}, \frac{3\pi}{8}\right)$ (D) $\left(2\sec\frac{3\pi}{8}, \frac{5\pi}{8}\right)$

Q48. JEE Main 2024 (Online) 29th January Morning Shift

If $z = \frac{1}{2} - 2i$ is such that $|z+1| = \alpha z + \beta(1+i), i = \sqrt{-1}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha + \beta$ is equal to

(A) 2 (B) -4 (C) 3 (D) -1

Q49. JEE Main 2024 (Online) 27th January Morning Shift

If $S = \{z \in C : |z-i| = |z+i| = |z-1|\}$, then, $n(S)$ is :

(A) 1 (B) 2 (C) 3 (D) 0

Q50. JEE Main 2023 (Online) 15th April Morning Shift

If the set $\left\{\mathrm{Re}\left(\frac{z-\bar{z}+z\bar{z}}{2-3z+5\bar{z}}\right) : z \in \mathbb{C}, \mathrm{Re}(z) = 3\right\}$ is equal to the interval $(\alpha, \beta]$, then $24(\beta - \alpha)$ is equal to :

(A) 36 (B) 27 (C) 42 (D) 30

Q51. JEE Main 2023 (Online) 13th April Evening Shift

Let $S = \left\{z \in \mathbb{C} : \bar{z} = i\left(z^2 + \mathrm{Re}(\bar{z})\right)\right\}$. Then $\sum\limits_{z \in \mathrm{S}} |z|^2$ is equal to :

(A) $\frac{7}{2}$ (B) 4 (C) 3 (D) $\frac{5}{2}$

Q52. JEE Main 2023 (Online) 12th April Morning Shift

Let C be the circle in the complex plane with centre $\mathrm{z}_0 = \frac{1}{2}(1+3i)$ and radius $r = 1$. Let $\mathrm{z}_1 = 1 + \mathrm{i}$ and the complex number z_2 be outside the circle C such that $|z_1 - z_0|\,|z_2 - z_0| = 1$. If z_0, z_1 and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to :

(A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{13}{2}$ (D) $\frac{7}{2}$

Q53. JEE Main 2023 (Online) 11th April Evening Shift

For $a \in \mathbb{C}$, let $\mathrm{A} = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) > \mathrm{Im}(\bar{a} + z)\}$ and $\mathrm{B} = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) < \mathrm{Im}(\bar{a} + z)\}$. Then among the two statements :

(S1): If $\mathrm{Re}(a), \mathrm{Im}(a) > 0$, then the set A contains all the real numbers

(S2) : If $\mathrm{Re}(a), \mathrm{Im}(a) < 0$, then the set B contains all the real numbers,

(A) both are false (B) only (S1) is true

(C) only (S2) is true (D) both are true

Q54. JEE Main 2023 (Online) 11th April Morning Shift

Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument of $w_1 - w_2$ is equal to :

(A) $-\pi + \tan^{-1}\frac{8}{9}$ (B) $-\pi + \tan^{-1}\frac{33}{5}$ (C) $\pi - \tan^{-1}\frac{8}{9}$ (D) $\pi - \tan^{-1}\frac{33}{5}$

Q55. JEE Main 2023 (Online) 10th April Evening Shift

Let $S = \left\{z = x + iy : \frac{2z-3i}{4z+2i} \text{ is a real number}\right\}$. Then which of the following is NOT correct?

(A) $y + x^2 + y^2 \neq -\frac{1}{4}$

(B) $(x, y) = \left(0, -\frac{1}{2}\right)$

(C) $x = 0$

(D) $y \in \left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, \infty\right)$

Q56. JEE Main 2023 (Online) 10th April Morning Shift

Let the complex number $z = x + iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to :

(A) $\frac{4}{3}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$

Q57. JEE Main 2023 (Online) 8th April Evening Shift

Let $A = \left\{\theta \in (0, 2\pi) : \frac{1+2i\sin\theta}{1-i\sin\theta} \text{ is purely imaginary }\right\}$. Then the sum of the elements in A is :

(A) 3π (B) π (C) 2π (D) 4π

Q58. JEE Main 2023 (Online) 8th April Morning Shift

If for $z = \alpha + i\beta$, $|z + 2| = z + 4(1 + i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation :

(A) $x^2 + 2x - 3 = 0$ (B) $x^2 + 3x - 4 = 0$ (C) $x^2 + x - 12 = 0$ (D) $x^2 + 7x + 12 = 0$

Q59. JEE Main 2023 (Online) 6th April Evening Shift

Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in \mathbb{C} : \text{Re}\left(az^2 + bz\right) = a \text{ and } \text{Re}\left(bz^2 + az\right) = b\right\}$ is equal to :

(A) 0 (B) 2 (C) 1 (D) Infinite

Q60. JEE Main 2023 (Online) 1st February Evening Shift

Let a, b be two real numbers such that $ab < 0$. If the complex number $\frac{1+ai}{b+i}$ is of unit modulus and $a + ib$ lies on the circle $|z - 1| = |2z|$, then a possible value of $\frac{1+[a]}{4b}$, where $[t]$ is greatest integer function, is :

(A) $\left(\frac{1+\sqrt{7}}{4}\right)$ (B) $\frac{1}{2}$

(C) 0 (D) -1

Q61. JEE Main 2023 (Online) 1st February Morning Shift

If the center and radius of the circle $\left|\frac{z-2}{z-3}\right| = 2$ are respectively (α, β) and γ, then $3(\alpha + \beta + \gamma)$ is equal to :

(A) 12 (B) 10 (C) 11 (D) 9

Q62. JEE Main 2023 (Online) 31st January Evening Shift

The complex number $z = \frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$ is equal to :

(A) $\cos\frac{\pi}{12} - i\sin\frac{\pi}{12}$ (B) $\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$

(C) $\sqrt{2}i\left(\cos\frac{5\pi}{12} - i\sin\frac{5\pi}{12}\right)$ (D) $\sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$

Q63. JEE Main 2023 (Online) 31st January Morning Shift

For all $z \in C$ on the curve $C_1 : |z| = 4$, let the locus of the point $z + \frac{1}{z}$ be the curve C_2. Then :

(A) the curves C_1 and C_2 intersect at 4 points

(B) the curve C_2 lies inside C_1

(C) the curve C_1 lies inside C_2

(D) the curves C_1 and C_2 intersect at 2 points

Q64. JEE Main 2023 (Online) 29th January Morning Shift

For two non-zero complex numbers z_1 and z_2, if $\mathrm{Re}(z_1 z_2) = 0$ and $\mathrm{Re}(z_1 + z_2) = 0$, then which of the following are possible?

A. $\mathrm{Im}(z_1) > 0$ and $\mathrm{Im}(z_2) > 0$

B. $\mathrm{Im}(z_1) < 0$ and $\mathrm{Im}(z_2) > 0$

C. $\mathrm{Im}(z_1) > 0$ and $\mathrm{Im}(z_2) < 0$

D. $\mathrm{Im}(z_1) < 0$ and $\mathrm{Im}(z_2) < 0$

Choose the correct answer from the options given below :

(A) A and C (B) A and B (C) B and D (D) B and C

Q65. JEE Main 2023 (Online) 25th January Evening Shift

Let z be a complex number such that $\left|\frac{z-2i}{z+i}\right| = 2, z \neq -i$. Then z lies on the circle of radius 2 and centre :

(A) $(0, -2)$ (B) $(0, 0)$ (C) $(0, 2)$ (D) $(2, 0)$

Q66. JEE Main 2023 (Online) 25th January Morning Shift

Let $z_1 = 2 + 3i$ and $z_2 = 3 + 4i$. The set $S = \left\{z \in \mathbb{C} : |z - z_1|^2 - |z - z_2|^2 = |z_1 - z_2|^2\right\}$ represents a

(A) hyperbola with the length of the transverse axis 7

(B) hyperbola with eccentricity 2

(C) straight line with the sum of its intercepts on the coordinate axes equals -18

(D) straight line with the sum of its intercepts on the coordinate axes equals 14

Q67. JEE Main 2023 (Online) 24th January Evening Shift

The value of $\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is

(A) $-\frac{1}{2}\left(1 - i\sqrt{3}\right)$ (B) $-\frac{1}{2}\left(\sqrt{3} - i\right)$

(C) $\frac{1}{2}\left(1 - i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(\sqrt{3} + i\right)$

Q68. JEE Main 2023 (Online) 24th January Morning Shift

Let $p, q \in \mathbb{R}$ and $\left(1 - \sqrt{3}i\right)^{200} = 2^{199}(p + iq), i = \sqrt{-1}$ then $p + q + q^2$ and $p - q + q^2$ are roots of the equation.

(A) $x^2 + 4x - 1 = 0$ (B) $x^2 - 4x + 1 = 0$ (C) $x^2 + 4x + 1 = 0$ (D) $x^2 - 4x - 1 = 0$

Q69. JEE Main 2022 (Online) 29th July Evening Shift

If $z \neq 0$ be a complex number such that $\left|z - \frac{1}{z}\right| = 2$, then the maximum value of $|z|$ is :

(A) $\sqrt{2}$ (B) 1 (C) $\sqrt{2} - 1$ (D) $\sqrt{2} + 1$

Q70. JEE Main 2022 (Online) 29th July Evening Shift

Let $S = \{z = x + iy : |z - 1 + i| \geq |z|, |z| < 2, |z + i| = |z - 1|\}$. Then the set of all values of x, for which $w = 2x + iy \in S$ for some $y \in \mathbb{R}$, is :

(A) $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$ (B) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$

(C) $\left(-\sqrt{2}, \frac{1}{2}\right]$ (D) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Q71. JEE Main 2022 (Online) 29th July Morning Shift

If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to :

(A) 244 (B) 224 (C) 245 (D) 265

Q72. JEE Main 2022 (Online) 28th July Morning Shift

Let $S_1 = \{z_1 \in \mathbf{C} : |z_1 - 3| = \frac{1}{2}\}$ and $S_2 = \{z_2 \in \mathbf{C} : |z_2 - |z_2 + 1|| = |z_2 + |z_2 - 1||\}$. Then, for $z_1 \in S_1$ and $z_2 \in S_2$, the least value of $|z_2 - z_1|$ is :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Q73. JEE Main 2022 (Online) 27th July Evening Shift

Let S be the set of all $(\alpha, \beta), \pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1 - i\sin\alpha}{1 + 2i\sin\alpha}$ is purely imaginary and $\frac{1 + i\cos\beta}{1 - 2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta, (\alpha, \beta) \in S$. Then $\sum\limits_{(\alpha,\beta)\in S} \left(iZ_{\alpha\beta} + \frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to :

(A) 3 (B) 3 i (C) 1 (D) 2 – i

Q74. JEE Main 2022 (Online) 27th July Morning Shift

Let the minimum value v_0 of $v = |z|^2 + |z - 3|^2 + |z - 6i|^2, z \in \mathbb{C}$ is attained at $z = z_0$. Then $\left|2z_0^2 - \bar{z}_0^3 + 3\right|^2 + v_0^2$ is equal to :

(A) 1000 (B) 1024 (C) 1105 (D) 1196

Q75. JEE Main 2022 (Online) 26th July Evening Shift

If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then :

(A) $x + 2y - 4 = 0$ (B) $x^2 + y - 4 = 0$ (C) $x + 2y + 4 = 0$ (D) $x^2 - y + 3 = 0$

Q76. JEE Main 2022 (Online) 26th July Morning Shift

Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $\mathrm{Re}(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

(A) $\arg z_2 = \pi - \tan^{-1}3$ (B) $\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$

(C) $|z_2| = \sqrt{10}$ (D) $|2z_1 - z_2| = 5$

Q77. JEE Main 2022 (Online) 25th July Evening Shift

For $z \in \mathbb{C}$ if the minimum value of $(|z - 3\sqrt{2}| + |z - p\sqrt{2}i|)$ is $5\sqrt{2}$, then a value of p is _____.

(A) 3 (B) $\frac{7}{2}$ (C) 4 (D) $\frac{9}{2}$

Q78. JEE Main 2022 (Online) 25th July Morning Shift

For $n \in \mathbf{N}$, let $S_n = \left\{z \in \mathbf{C} : |z - 3 + 2i| = \frac{n}{4}\right\}$ and $T_n = \left\{z \in \mathbf{C} : |z - 2 + 3i| = \frac{1}{n}\right\}$. Then the number of elements in the set $\{n \in \mathbf{N} : S_n \cap T_n = \phi\}$ is :

(A) 0 (B) 2 (C) 3 (D) 4

Q79. JEE Main 2022 (Online) 30th June Morning Shift

The real part of the complex number $\frac{(1+2i)^8 \cdot (1-2i)^2}{(3+2i) \cdot \overline{(4-6i)}}$ is equal to :

(A) $\frac{500}{13}$ (B) $\frac{110}{13}$ (C) $\frac{55}{6}$ (D) $\frac{550}{13}$

Q80. JEE Main 2022 (Online) 29th June Evening Shift

Let arg(z) represent the principal argument of the complex number z. Then, |z| = 3 and arg(z − 1) − arg(z + 1) = $\frac{\pi}{4}$ intersect :

(A) exactly at one point. (B) exactly at two points.

(C) nowhere. (D) at infinitely many points.

Q81. JEE Main 2022 (Online) 29th June Morning Shift

Let α and β be the roots of the equation $x^2 + (2i - 1) = 0$. Then, the value of $|\alpha^8 + \beta^8|$ is equal to :

(A) 50 (B) 250 (C) 1250 (D) 1500

Q82. JEE Main 2022 (Online) 27th June Evening Shift

The number of points of intersection of

$|z - (4 + 3i)| = 2$ and $|z| + |z - 4| = 6$, $z \in C$, is :

(A) 0 (B) 1 (C) 2 (D) 3

Q83. JEE Main 2022 (Online) 27th June Morning Shift

The area of the polygon, whose vertices are the non-real roots of the equation $\overline{z} = iz^2$ is :

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3}{2}$ (D) $\frac{3}{4}$

Q84. JEE Main 2022 (Online) 26th June Morning Shift

Let $A = \left\{z \in C : \left|\frac{z+1}{z-1}\right| < 1\right\}$ and $B = \left\{z \in C : \arg\left(\frac{z-1}{z+1}\right) = \frac{2\pi}{3}\right\}$. Then A ∩ B is :

(A) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second and third quadrants only

(B) a portion of a circle centred at $\left(0, -\frac{1}{\sqrt{3}}\right)$ that lies in the second quadrant only

(C) an empty

(D) a portion of a circle of radius $\frac{2}{\sqrt{3}}$ that lies in the third quadrant only

Q85. JEE Main 2022 (Online) 25th June Evening Shift

Let z_1 and z_2 be two complex numbers such that $\overline{z}_1 = i\overline{z}_2$ and $\arg\left(\frac{z_1}{\overline{z}_2}\right) = \pi$. Then :

(A) $\arg z_2 = \frac{\pi}{4}$

(B) $\arg z_2 = -\frac{3\pi}{4}$

(C) $\arg z_1 = \frac{\pi}{4}$

(D) $\arg z_1 = -\frac{3\pi}{4}$

Q86. JEE Main 2022 (Online) 25th June Morning Shift

Let a circle C in complex plane pass through the points $z_1 = 3 + 4i$, $z_2 = 4 + 3i$ and $z_3 = 5i$. If $z(\neq z_1)$ is a point on C such that the line through z and z_1 is perpendicular to the line through z_2 and z_3, then $arg(z)$ is equal to :

(A) $\tan^{-1}\left(\frac{2}{\sqrt{5}}\right) - \pi$

(B) $\tan^{-1}\left(\frac{24}{7}\right) - \pi$

(C) $\tan^{-1}(3) - \pi$

(D) $\tan^{-1}\left(\frac{3}{4}\right) - \pi$

Q87. JEE Main 2022 (Online) 24th June Morning Shift

Let $A = \{z \in C : 1 \leq |z - (1+i)| \leq 2\}$

and $B = \{z \in A : |z - (1-i)| = 1\}$. Then, B :

(A) is an empty set

(B) contains exactly two elements

(C) contains exactly three elements

(D) is an infinite set

Q88. JEE Main 2021 (Online) 31st August Evening Shift

If z is a complex number such that $\frac{z-i}{z-1}$ is purely imaginary, then the minimum value of | z − (3 + 3i) | is :

(A) $2\sqrt{2} - 1$

(B) $3\sqrt{2}$

(C) $6\sqrt{2}$

(D) $2\sqrt{2}$

Q89. JEE Main 2021 (Online) 27th August Morning Shift

If $S = \left\{z \in C : \frac{z-i}{z+2i} \in R\right\}$, then :

(A) S contains exactly two elements

(B) S contains only one element

(C) S is a circle in the complex plane

(D) S is a straight line in the complex plane

Q90. JEE Main 2021 (Online) 26th August Evening Shift

If $\left(\sqrt{3} + i\right)^{100} = 2^{99}(p + iq)$, then p and q are roots of the equation :

(A) $x^2 - \left(\sqrt{3} - 1\right)x - \sqrt{3} = 0$

(B) $x^2 + \left(\sqrt{3} + 1\right)x + \sqrt{3} = 0$

(C) $x^2 + \left(\sqrt{3} - 1\right)x - \sqrt{3} = 0$

(D) $x^2 - \left(\sqrt{3} + 1\right)x + \sqrt{3} = 0$

Q91. JEE Main 2021 (Online) 26th August Morning Shift

The equation $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ represents a circle with :

(A) centre at (0, -1) and radius $\sqrt{2}$

(B) centre at (0, 1) and radius $\sqrt{2}$

(C) centre (0, 0) and radius $\sqrt{2}$

(D) centre at (0, 1) and radius 2

Q92. JEE Main 2021 (Online) 27th July Evening Shift

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z - 2| \leq 1\}$ and

$S_2 = \{z \in C : z(1 + i) + \overline{z}(1 - i) \geq 4\}$.

Then, the maximum value of $\left|z - \frac{5}{2}\right|^2$ for $z \in S_1 \cap S_2$ is equal to :

(A) $\frac{3+2\sqrt{2}}{4}$ (B) $\frac{5+2\sqrt{2}}{2}$ (C) $\frac{3+2\sqrt{2}}{2}$ (D) $\frac{5+2\sqrt{2}}{4}$

Q93. JEE Main 2021 (Online) 27th July Morning Shift

Let C be the set of all complex numbers. Let

$S_1 = \{z \in C : |z - 3 - 2i|^2 = 8\}$

$S_2 = \{z \in C : \text{Re}(z) \geq 5\}$ and

$S_3 = \{z \in C : |z - \overline{z}| \geq 8\}$.

Then the number of elements in $S_1 \cap S_2 \cap S_3$ is equal to :

(A) 1 (B) 0 (C) 2 (D) Infinite

Q94. JEE Main 2021 (Online) 22th July Evening Shift

Let n denote the number of solutions of the equation $z^2 + 3\overline{z} = 0$, where z is a complex number.
Then the value of $\sum_{k=0}^{\infty} \frac{1}{n^k}$ is equal to :

(A) 1 (B) $\frac{4}{3}$ (C) $\frac{3}{2}$ (D) 2

Q95. JEE Main 2021 (Online) 20th July Morning Shift

If z and ω are two complex numbers such that $|z\omega| = 1$ and $\arg(z) - \arg(\omega) = \frac{3\pi}{2}$, then $\arg\left(\frac{1-2\bar{z}\omega}{1+3\bar{z}\omega}\right)$ is :

(Here arg(z) denotes the principal argument of complex number z)

(A) $\frac{\pi}{4}$ (B) $-\frac{3\pi}{4}$ (C) $-\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

Q96. JEE Main 2021 (Online) 18th March Evening Shift

Let a complex number be w = $1-\sqrt{3}$i. Let another complex number z be such that |zw| = 1 and arg(z) $-$ arg(w) = $\frac{\pi}{2}$. Then the area of the triangle with vertices origin, z and w is equal to :

(A) 4 (B) $\frac{1}{4}$ (C) 2 (D) $\frac{1}{2}$

Q97. JEE Main 2021 (Online) 18th March Morning Shift

If the equation $a|z|^2+\overline{\overline{\alpha}z+\alpha\overline{z}}+d=0$ represents a circle where a, d are real constants then which of the following condition is correct?

(A) $|\alpha|^2-ad\neq 0$ (B) $|\alpha|^2-ad>0$ and $a\in R-\{0\}$

(C) $|\alpha|^2-ad\geq 0$ and $a\in R$ (D) $\alpha=0$, $a, d\in R^+$

Q98. JEE Main 2021 (Online) 17th March Evening Shift

Let S_1, S_2 and S_3 be three sets defined as

$S_1=\{z\in C : |z-1|\leq\sqrt{2}\}$

$S_2=\{z\in C : Re((1-i)z)\geq 1\}$

$S_3=\{z\in C : Im(z)\leq 1\}$

Then the set $S_1\cap S_2\cap S_3$:

(A) has exactly three elements (B) is a singleton

(C) has infinitely many elements (D) has exactly two elements

Q99. JEE Main 2021 (Online) 17th March Morning Shift

The area of the triangle with vertices A(z), B(iz) and C(z + iz) is :

(A) 1 (B) $\frac{1}{2}|z|^2$

(C) $\frac{1}{2}|z+iz|^2$ (D) $\frac{1}{2}$

Q100. JEE Main 2021 (Online) 16th March Evening Shift

The least value of |z| where z is complex number which satisfies the inequality $\exp\left(\frac{(|z|+3)(|z|-1)}{||z|+1|}\log_e 2\right)\geq\log_{\sqrt{2}}|5\sqrt{7}+9i|, i=\sqrt{-1}$, is equal to :

(A) 8 (B) 3 (C) 2 (D) $\sqrt{5}$

Q101. JEE Main 2021 (Online) 16th March Morning Shift

Let a complex number z, $|z| \neq 1$,

satisfy $\log_{\frac{1}{\sqrt{2}}}\left(\frac{|z|+11}{(|z|-1)^2}\right) \leq 2$. Then, the largest value of |z| is equal to _____.

(A) 5 (B) 8 (C) 6 (D) 7

Q102. JEE Main 2021 (Online) 25th February Evening Shift

If $\alpha, \beta \in$ R are such that $1 - 2i$ (here $i^2 = -1$) is a root of $z^2 + \alpha z + \beta = 0$, then $(\alpha - \beta)$ is equal to :

(A) -7 (B) 7 (C) 3 (D) -3

Q103. JEE Main 2021 (Online) 25th February Morning Shift

Let the lines $(2 - i)z = (2 + i)\overline{z}$ and $(2 + i)z + (i - 2)\overline{z} - 4i = 0$, (here $i^2 = -1$) be normal to a circle C. If the line $iz + \overline{z} + 1 + i = 0$ is tangent to this circle C, then its radius is :

(A) $\frac{3}{2\sqrt{2}}$ (B) $3\sqrt{2}$ (C) $\frac{1}{2\sqrt{2}}$ (D) $\frac{3}{\sqrt{2}}$

Q104. JEE Main 2020 (Online) 6th September Evening Slot

Let z = x + iy be a non-zero complex number such that $z^2 = i|z|^2$, where $i = \sqrt{-1}$, then z lies on the :

(A) line, y = −x (B) real axis (C) line, y = x (D) imaginary axis

Q105. JEE Main 2020 (Online) 6th September Morning Slot

The region represented by
$\{z = x + iy \in C : |z| - Re(z) \leq 1\}$ is also given by the inequality :

(A) $y^2 \leq 2\left(x + \frac{1}{2}\right)$ (B) $y^2 \leq x + \frac{1}{2}$

(C) $y^2 \geq 2(x + 1)$ (D) $y^2 \geq x + 1$

Q106. JEE Main 2020 (Online) 5th September Evening Slot

The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is :

(A) $-2^{15}i$ (B) -2^{15} (C) $2^{15}i$ (D) 6^5

Q107. JEE Main 2020 (Online) 5th September Morning Slot

If the four complex numbers $z, \overline{z}, \overline{z} - 2\mathrm{Re}(\overline{z})$ and $z - 2Re(z)$ represent the vertices of a square of side 4 units in the Argand plane, then $|z|$ is equal to :

(A) $4\sqrt{2}$ (B) 4 (C) 2 (D) $2\sqrt{2}$

Q108. JEE Main 2020 (Online) 4th September Evening Slot

If a and b are real numbers such that
$(2+\alpha)^4 = a + b\alpha$
where $\alpha = \frac{-1+i\sqrt{3}}{2}$ then a + b is equal to :

(A) 33 (B) 9 (C) 24 (D) 57

Q109. JEE Main 2020 (Online) 4th September Morning Slot

Let $u = \frac{2z+i}{z-ki}$, z = x + iy and k > 0. If the curve represented
by Re(u) + Im(u) = 1 intersects the y-axis at the points P and Q where PQ = 5, then the value of k is :

(A) 2 (B) 4 (C) 1/2 (D) 3/2

Q110. JEE Main 2020 (Online) 3rd September Evening Slot

If z_1 , z_2 are complex numbers such that
$Re(z_1) = |z_1 - 1|$, $Re(z_2) = |z_2 - 1|$, and
$arg(z_1 - z_2) = \frac{\pi}{6}$, then $Im(z_1 + z_2)$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}}$ (D) $2\sqrt{3}$

Q111. JEE Main 2020 (Online) 2nd September Evening Slot

The imaginary part of
$\left(3+2\sqrt{-54}\right)^{\frac{1}{2}} - \left(3-2\sqrt{-54}\right)^{\frac{1}{2}}$ can be :

(A) $-2\sqrt{6}$ (B) 6 (C) $\sqrt{6}$ (D) $-\sqrt{6}$

Q112. JEE Main 2020 (Online) 2nd September Morning Slot

The value of

$\left(\frac{1+\sin\frac{2\pi}{9}+i\cos\frac{2\pi}{9}}{1+\sin\frac{2\pi}{9}-i\cos\frac{2\pi}{9}}\right)^3$ is :

(A) $\frac{1}{2}\left(\sqrt{3}-i\right)$ (B) $-\frac{1}{2}\left(\sqrt{3}-i\right)$

(C) $-\frac{1}{2}\left(1-i\sqrt{3}\right)$ (D) $\frac{1}{2}\left(1-i\sqrt{3}\right)$

Q113. JEE Main 2020 (Online) 9th January Evening Slot

If z be a complex number satisfying |Re(z)| + |Im(z)| = 4, then |z| cannot be :

(A) $\sqrt{10}$ (B) $\sqrt{7}$ (C) $\sqrt{\frac{17}{2}}$ (D) $\sqrt{8}$

Q114. JEE Main 2020 (Online) 9th January Morning Slot

Let z be complex number such that
$\left|\frac{z-i}{z+2i}\right| = 1$ and |z| = $\frac{5}{2}$.
Then the value of |z + 3i| is :

(A) $2\sqrt{3}$ (B) $\sqrt{10}$ (C) $\frac{15}{4}$ (D) $\frac{7}{2}$

Q115. JEE Main 2020 (Online) 8th January Morning Slot

If the equation, x^2 + bx + 45 = 0 (b $\in$ R) has conjugate complex roots and they satisfy |z +1| = $2\sqrt{10}$, then :

(A) b^2 – b = 42 (B) b^2 + b = 12 (C) b^2 + b = 72 (D) b^2 – b = 30

Q116. JEE Main 2020 (Online) 7th January Evening Slot

If $\frac{3+i\sin\theta}{4-i\cos\theta}$, $\theta \in [0, 2\pi]$, is a real number, then an argument of
sinθ + icosθ is :

(A) $\pi - \tan^{-1}\left(\frac{3}{4}\right)$ (B) $-\tan^{-1}\left(\frac{3}{4}\right)$
(C) $\tan^{-1}\left(\frac{4}{3}\right)$ (D) $\pi - \tan^{-1}\left(\frac{4}{3}\right)$

Q117. JEE Main 2020 (Online) 7th January Morning Slot

If $\mathrm{Re}\left(\frac{z-1}{2z+i}\right) = 1$, where z = x + iy, then the point (x, y) lies on a :

(A) straight line whose slope is $\frac{3}{2}$
(B) straight line whose slope is $-\frac{2}{3}$
(C) circle whose diameter is $\frac{\sqrt{5}}{2}$
(D) circle whose centre is at $\left(-\frac{1}{2}, -\frac{3}{2}\right)$

Q118. JEE Main 2019 (Online) 12th April Evening Slot

Let z $\in$ C with Im(z) = 10 and it satisfies $\frac{2z-n}{2z+n}$ = 2i - 1 for some natural number n. Then :

(A) n = 20 and Re(z) = –10 (B) n = 40 and Re(z) = 10
(C) n = 40 and Re(z) = –10 (D) n = 20 and Re(z) = 10

Q119. JEE Main 2019 (Online) 12th April Morning Slot

The equation |z – i| = |z – 1|, i = $\sqrt{-1}$, represents :

(A) a circle of radius 1
(B) the line through the origin with slope – 1
(C) a circle of radius $\frac{1}{2}$
(D) the line through the origin with slope 1

Q120. JEE Main 2019 (Online) 10th April Evening Slot

If z and w are two complex numbers such that |zw| = 1 and arg(z) − arg(w) = $\frac{\pi}{2}$, then :

(A) $z\overline{w} = \frac{1-i}{\sqrt{2}}$ (B) $\overline{z}w = i$

(C) $z\overline{w} = \frac{-1+i}{\sqrt{2}}$ (D) $\overline{z}w = -i$

Q121. JEE Main 2019 (Online) 10th April Morning Slot

If a > 0 and z = $\frac{(1+i)^2}{a-i}$, has magnitude $\sqrt{\frac{2}{5}}$, then $\overline{z}$ is equal to :

(A) $-\frac{1}{5} + \frac{3}{5}i$ (B) $-\frac{1}{5} - \frac{3}{5}i$ (C) $\frac{1}{5} - \frac{3}{5}i$ (D) $-\frac{3}{5} - \frac{1}{5}i$

Q122. JEE Main 2019 (Online) 9th April Evening Slot

Let z ∈ C be such that |z| < 1.

If $\omega = \frac{5+3z}{5(1-z)}$, then :

(A) 4Im(ω) > 5 (B) 5Im(ω) < 1 (C) 5Re(ω) > 4 (D) 5Re(ω) > 1

Q123. JEE Main 2019 (Online) 9th April Morning Slot

All the points in the set
$S = \left\{\frac{\alpha+i}{\alpha-i} : \alpha \in R\right\}(i = \sqrt{-1})$ lie on a :

(A) straight line whose slope is −1 (B) straight line whose slope is 1.

(C) circle whose radius is 1. (D) circle whose radius is $\sqrt{2}$.

Q124. JEE Main 2019 (Online) 8th April Evening Slot

If $z = \frac{\sqrt{3}}{2} + \frac{i}{2}\left(i = \sqrt{-1}\right)$,

then $(1 + iz + z^5 + iz^8)^9$ is equal to :

(A) 1 (B) −1 (C) 0 (D) $(-1 + 2i)^9$

Q125. JEE Main 2019 (Online) 8th April Morning Slot

If α and β be the roots of the equation $x^2 - 2x + 2 = 0$, then the least value of n for which $\left(\frac{\alpha}{\beta}\right)^n = 1$ is :

(A) 2 (B) 5 (C) 4 (D) 3

Q126. JEE Main 2019 (Online) 12th January Evening Slot

Let z_1 and z_2 be two complex numbers satisfying $|z_1| = 9$ and $|z_2 - 3 - 4i| = 4$. Then the minimum value of $|z_1 - z_2|$ is :

(A) 0 (B) 1 (C) 2 (D) $\sqrt{2}$

Q127. JEE Main 2019 (Online) 12th January Morning Slot

If $\frac{z-\alpha}{z+\alpha}$ $(\alpha \in R)$ is a purely imaginary number and | z | = 2, then a value of α is :

(A) $\frac{1}{2}$ (B) $\sqrt{2}$ (C) 2 (D) 1

Q128. JEE Main 2019 (Online) 11th January Evening Slot

Let z be a complex number such that |z| + z = 3 + i (where i = $\sqrt{-1}$). Then |z| is equal to :

(A) $\frac{\sqrt{34}}{3}$ (B) $\frac{5}{3}$ (C) $\frac{5}{4}$ (D) $\frac{\sqrt{41}}{4}$

Q129. JEE Main 2019 (Online) 11th January Morning Slot

Let $\left(-2-\frac{1}{3}i\right)^3=\frac{x+iy}{27}\left(i=\sqrt{-1}\right)$, where x and y are real numbers, then y $-$ x equals :

(A) $-$ 85 (B) 85 (C) $-$ 91 (D) 91

Q130. JEE Main 2019 (Online) 10th January Evening Slot

Let $z=\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5+\left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5$. If R(z) and I(z) respectively denote the real and imaginary parts of z, then :

(A) R(z) = $-$ 3 (B) R(z) < 0 and I(z) > 0

(C) I(z) = 0 (D) R(z) > 0 and I(z) > 0

Q131. JEE Main 2019 (Online) 10th January Morning Slot

Let z_1 and z_2 be any two non-zero complex numbers such that $3|z_1|=4|z_2|$. If $z=\frac{3z_1}{2z_2}+\frac{2z_2}{3z_1}$ then :

(A) $Im(z)=0$ (B) $|z|=\sqrt{\frac{17}{2}}$

(C) $|z|=\frac{1}{2}\sqrt{9+16\cos^2\theta}$ (D) Re(z) $=0$

Q132. JEE Main 2019 (Online) 9th January Evening Slot

Let z_0 be a root of the quadratic equation, $x^2 + x + 1 = 0$, If $z = 3 + 6iz_0^{81} - 3iz_0^{93}$, then arg z is equal to :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{3}$ (D) 0

Q133. JEE Main 2019 (Online) 9th January Morning Slot

Let α and β be two roots of the equation $x^2 + 2x + 2 = 0$, then $\alpha^{15} + \beta^{15}$ is equal to :

(A) -256 (B) 512 (C) -512 (D) 256

Q134. JEE Main 2019 (Online) 9th January Morning Slot

Let
A = $\left\{\theta \in \left(-\frac{\pi}{2}, \pi\right) : \frac{3+2i\sin\theta}{1-2i\sin\theta} is\, purely\, imaginary\right\}$
. Then the sum of the elements in A is :

(A) $\frac{5\pi}{6}$ (B) π (C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$

Q135. JEE Main 2018 (Online) 16th April Morning Slot

The least positive integer n for which $\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^n = 1$, is :

(A) 2 (B) 3 (C) 5 (D) 6

Q136. JEE Main 2018 (Offline)

If $\alpha, \beta \in C$ are the distinct roots of the equation
$x^2 - x + 1 = 0$, then $\alpha^{101} + \beta^{107}$ is equal to :

(A) 2 (B) -1 (C) 0 (D) 1

Q137. JEE Main 2018 (Online) 15th April Evening Slot

If $|z - 3 + 2i| \leq 4$ then the difference between the greatest value and the least value of |z| is :

(A) $2\sqrt{13}$ (B) 8 (C) $4 + \sqrt{13}$ (D) $\sqrt{13}$

Q138. JEE Main 2018 (Online) 15th April Morning Slot

The set of all $\alpha \in$ **R**, for which $w = \frac{1+(1-8\alpha)z}{1-z}$ is purely imaginary number, for all $z \in$ **C** satisfying $|z| = 1$ and Re $z \neq 1$, is :

(A) an empty set (B) {0}
(C) $\left\{0, \frac{1}{4}, -\frac{1}{4}\right\}$ (D) equal to **R**

Q139. JEE Main 2017 (Online) 9th April Morning Slot

The equation
$\text{Im}\left(\frac{iz-2}{z-i}\right) + 1 = 0, z \in$ **C**, $z \neq i$
represents a part of a circle having radius equal to :

(A) 2 (B) 1 (C) $\frac{3}{4}$ (D) $\frac{1}{2}$

Q140. JEE Main 2017 (Online) 8th April Morning Slot

Let z∈C, the set of complex numbers. Then the equation, $2|z + 3i| - |z - i| = 0$ represents :

(A) a circle with radius $\frac{8}{3}$.

(B) a circle with diameter $\frac{10}{3}$.

(C) an ellipse with length of major axis $\frac{16}{3}$.

(D) an ellipse with length of minor axis $\frac{16}{9}$.

Q141. JEE Main 2017 (Offline)

Let ω be a complex number such that $2\omega + 1 = z$ where $z = \sqrt{-3}$. If

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k,$$

then k is equal to :

(A) z (B) -1 (C) 1 (D) -z

Q142. JEE Main 2016 (Online) 9th April Morning Slot

The point represented by 2 + *i* in the Argand plane moves 1 unit eastwards, then 2 units northwards and finally from there $2\sqrt{2}$ units in the south-westwardsdirection. Then its new position in the Argand plane is at the point represented by :

(A) 2 + 2i (B) 1 + i (C) $-1 - i$ (D) $-2 - 2i$

Q143. JEE Main 2016 (Offline)

A value of θ for which $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely imaginary, is :

(A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (B) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Q144. JEE Main 2015 (Offline)

A complex number z is said to be unimodular if $|z| = 1$. Suppose z_1 and z_2 are complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular and z_2 is not unimodular. Then the point z_1 lies on a :

(A) circle of radius 2. (B) circle of radius $\sqrt{2}$.

(C) straight line parallel to x-axis (D) straight line parallel to y-axis.

Q145. JEE Main 2014 (Offline)

If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$:

(A) is strictly greater that $\frac{5}{2}$

(B) is strictly greater that $\frac{3}{2}$ but less than $\frac{5}{2}$

(C) is equal to $\frac{5}{2}$

(D) lie in the interval (1, 2)

Q146. JEE Main 2013 (Offline)

If z is a complex number of unit modulus and argument θ, then arg $\left(\frac{1+z}{1+\bar{z}}\right)$ equals :

(A) $-\theta$ (B) $\frac{\pi}{2} - \theta$ (C) θ (D) $\pi - \theta$

Q147. AIEEE 2012

If $z \neq 1$ and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies :

(A) either on the real axis or a circle passing through the origin.

(B) on a circle with centre at the origin

(C) either on real axis or on a circle not passing through the origin.

(D) on the imaginary axis.

Q148. AIEEE 2011

Let α, β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line Re z = 1, then it is necessary that :

(A) $\beta \in (-1, 0)$ (B) $|\beta| = 1$ (C) $\beta \in (1, \infty)$ (D) $\beta \in (0, 1)$

Q149. AIEEE 2011

If $\omega(\neq 1)$ is a cube root of unity, and $(1 + \omega)^7 = A + B\omega$. Then (A, B) equals :

(A) (1 ,1) (B) (1, 0) (C) (- 1 ,1) (D) (0 ,1)

Q150. AIEEE 2010

The number of complex numbers z such that $|z - 1| = |z + 1| = |z - i|$ equals :

(A) 1 (B) 2 (C) ∞ (D) 0

Q151. AIEEE 2009

If $\left|z - \frac{4}{z}\right| = 2$, then the maximum value of $|z|$ is equal to :

(A) $\sqrt{5} + 1$ (B) 2 (C) $2 + \sqrt{2}$ (D) $\sqrt{3} + 1$

Q152. AIEEE 2008

The conjugate of a complex number is $\frac{1}{i-1}$ then that complex number is :

(A) $\frac{-1}{i-1}$ (B) $\frac{1}{i+1}$ (C) $\frac{-1}{i+1}$ (D) $\frac{1}{i-1}$

Q153. AIEEE 2007

If $|z+4| \leq 3$, then the maximum value of $|z+1|$ is :

(A) 6 (B) 0 (C) 4 (D) 10

Q154. AIEEE 2006

The value of $\sum_{k=1}^{10} \left(\sin \frac{2k\pi}{11} + i \cos \frac{2k\pi}{11}\right)$ is :

(A) i (B) 1 (C) - 1 (D) - i

Q155. AIEEE 2006

If $z^2 + z + 1 = 0$, where z is a complex number, then the value of $\left(z+\frac{1}{z}\right)^2 + \left(z^2+\frac{1}{z^2}\right)^2 + \left(z^3+\frac{1}{z^3}\right)^2 + \ldots\ldots\ldots + \left(z^6+\frac{1}{z^6}\right)^2$ is :

(A) 18 (B) 54 (C) 6 (D) 12

Q156. AIEEE 2005

If the cube roots of unity are 1, ω, ω^2 then the roots of the equation $(x-1)^3$ + 8 = 0, are :

(A) $-1, -1+2\,\omega, -1-2\,\omega^2$ (B) $-1, -1, -1$

(C) $-1, 1-2\omega, 1-2\omega^2$ (D) $-1, 1+2\omega, 1+2\omega^2$

Q157. AIEEE 2005

If z_1 and z_2 are two non-zero complex numbers such that $|z_1+z_2| = |z_1| + |z_2|$, then arg z_1 - arg z_2 is equal to :

(A) $\frac{\pi}{2}$ (B) $-\pi$ (C) 0 (D) $\frac{-\pi}{2}$

Q158. AIEEE 2005

If $\omega = \frac{z}{z-\frac{1}{3}i}$ and $|\omega| = 1$, then z lies on :

(A) an ellipse (B) a circle (C) a straight line (D) a parabola

Q159. AIEEE 2004

Let z and w be complex numbers such that $\overline{z} + i\overline{w} = 0$ and arg zw = π. Then arg z equals :

(A) $\frac{5\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) $\frac{\pi}{4}$

Q160. AIEEE 2004

If $z = x - iy$ and $z^{\frac{1}{3}} = p + iq$, then

$\frac{\left(\frac{x}{p}+\frac{y}{q}\right)}{(p^2+q^2)}$ is equal to :

(A) - 2 (B) - 1 (C) 2 (D) 1

Q161. AIEEE 2004

If $|z^2 - 1| = |z|^2 + 1$, then z lies on :

(A) an ellipse (B) the imaginary axis

(C) a circle (D) the real axis

Q162. AIEEE 2003

If z and ω are two non-zero complex numbers such that $|z\omega| = 1$ and $Arg(z) - Arg(\omega) = \frac{\pi}{2}$, then $\overline{z}\omega$ is equal to

(A) $-i$ (B) 1 (C) - 1 (D) i

Q163. AIEEE 2003

Let Z_1 and Z_2 be two roots of the equation $Z^2 + aZ + b = 0$, Z being complex. Further , assume that the origin, Z_1 and Z_2 form an equilateral triangle. Then :

(A) $a^2 = 4b$ (B) $a^2 = b$ (C) $a^2 = 2b$ (D) $a^2 = 3b$

Q164. AIEEE 2003

If $\left(\frac{1+i}{1-i}\right)^x = 1$ then :

(A) x = 2n + 1, where n is any positive integer

(B) x = 4n , where n is any positive integer

(C) x = 2n, where n is any positive integer

(D) x = 4n + 1, where n is any positive integer.

Q165. AIEEE 2002

z and w are two nonzero complex numbers such that $|z| = |w|$ and Arg z + Arg w =π then z equals

(A) $\overline{\omega}$ (B) $-\overline{\omega}$ (C) ω (D) $-\omega$

Q166. AIEEE 2002

If $|z - 4| < |z - 2|$, its solution is given by :

(A) $\mathrm{Re}(z) > 0$ (B) $\mathrm{Re}(z) < 0$

(C) $\mathrm{Re}(z) > 3$ (D) $\mathrm{Re}(z) > 2$

Q167. AIEEE 2002

The locus of the centre of a circle which touches the circle $|z - z_1| = a$ and $|z - z_2| = b$ externally

(z, z_1 & z_2 are complex numbers) will be :

(A) an ellipse (B) a hyperbola (C) a circle (D) none of these

Numerical 38 questions

Q168. JEE Main 2026 (Online) 24th January Evening Shift

Let $z = (1+i)(1+2i)(1+3i)\ldots(1+ni)$, where $i = \sqrt{-1}$. If $|z|^2 = 44200$, then n is equal to ____

Q169. JEE Main 2026 (Online) 22nd January Morning Shift

Let $\alpha = \frac{-1+i\sqrt{3}}{2}$ and $\beta = \frac{-1-i\sqrt{3}}{2}, i = \sqrt{-1}$. If

$(7-7\alpha+9\beta)^{20} + (9+7\alpha-7\beta)^{20} + (-7+9\alpha+7\beta)^{20} + (14+7\alpha+7\beta)^{20} = m^{10}$,

then m is ____

Q170. JEE Main 2025 (Online) 4th April Evening Shift

If α is a root of the equation $x^2 + x + 1 = 0$ and $\sum_{k=1}^{n} \left(\alpha^k + \frac{1}{\alpha^k}\right)^2 = 20$, then n is equal to _____.

Q171. JEE Main 2025 (Online) 4th April Morning Shift

Let $A = \{z \in C : |z - 2 - i| = 3\}, B = \{z \in C : Re(z - iz) = 2\}$ and $S = A \cap B$. Then $\sum_{z \in S} |z|^2$ is equal to _____.

Q172. JEE Main 2025 (Online) 29th January Evening Shift

Let integers $a, b \in [-3, 3]$ be such that $a + b \neq 0$. Then the number of all possible ordered pairs (a, b), for which $\left|\frac{z-a}{z+b}\right| = 1$ and $\begin{vmatrix} z+1 & \omega & \omega^2 \\ \omega & z+\omega^2 & 1 \\ \omega^2 & 1 & z+\omega \end{vmatrix} = 1, z \in C$, where ω and ω^2 are the roots of $x^2 + x + 1 = 0$, is equal to _____ .

Q173. JEE Main 2025 (Online) 23rd January Evening Shift

Let α, β be the roots of the equation $x^2 - ax - b = 0$ with $Im(\alpha) < Im(\beta)$. Let $P_n = \alpha^n - \beta^n$. If $P_3 = -5\sqrt{7}i, P_4 = -3\sqrt{7}i, P_5 = 11\sqrt{7}i$ and $P_6 = 45\sqrt{7}i$, then $|\alpha^4 + \beta^4|$ is equal to _____.

Q174. JEE Main 2024 (Online) 9th April Morning Shift

The sum of the square of the modulus of the elements in the set $\{z = \mathrm{a} + \mathrm{ib} : \mathrm{a}, \mathrm{b} \in \mathbf{Z}, z \in \mathbf{C}, |z-1| \leq 1, |z-5| \leq |z-5\mathrm{i}|\}$ is _____.

Q175. JEE Main 2024 (Online) 1st February Morning Shift

Let $\mathrm{P} = \{\mathrm{z} \in \mathbb{C} : |z+2-3i| \leq 1\}$ and $\mathrm{Q} = \{\mathrm{z} \in \mathbb{C} : z(1+i) + \bar{z}(1-i) \leq -8\}$. Let in $\mathrm{P} \cap \mathrm{Q}$, $|z-3+2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2 + 2|z_2|^2 = \alpha + \beta\sqrt{2}$, where α, β are integers, then $\alpha + \beta$ equals _____.

Q176. JEE Main 2024 (Online) 31st January Morning Shift

If α denotes the number of solutions of $|1-i|^x = 2^x$ and $\beta = \left(\frac{|z|}{\arg(z)}\right)$, where $z = \frac{\pi}{4}(1+i)^4 \left[\frac{1-\sqrt{\pi}i}{\sqrt{\pi}+i} + \frac{\sqrt{\pi}-i}{1+\sqrt{\pi}i}\right], i = \sqrt{-1}$, then the distance of the point (α, β) from the line $4x - 3y = 7$ is _____.

Q177. JEE Main 2024 (Online) 29th January Evening Shift

Let α, β be the roots of the equation $x^2 - \sqrt{6}x + 3 = 0$ such that $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Let a, b be integers not divisible by 3 and n be a natural number such that $\frac{\alpha^{99}}{\beta} + \alpha^{98} = 3^n(a+ib), i = \sqrt{-1}$. Then $n + a + b$ is equal to _____.

Q178. JEE Main 2024 (Online) 29th January Morning Shift

Let α, β be the roots of the equation $x^2 - x + 2 = 0$ with $\mathrm{Im}(\alpha) > \mathrm{Im}(\beta)$. Then $\alpha^6 + \alpha^4 + \beta^4 - 5\alpha^2$ is equal to _____.

Q179. JEE Main 2024 (Online) 27th January Evening Shift

Let the complex numbers α and $\frac{1}{\bar{\alpha}}$ lie on the circles $|z - z_0|^2 = 4$ and $|z - z_0|^2 = 16$ respectively, where $z_0 = 1 + i$. Then, the value of $100|\alpha|^2$ is _____.

Q180. JEE Main 2024 (Online) 27th January Morning Shift

If α satisfies the equation $x^2 + x + 1 = 0$ and $(1+\alpha)^7 = A + B\alpha + C\alpha^2, A, B, C \geqslant 0$, then $5(3A - 2B - C)$ is equal to _____.

Q181. JEE Main 2023 (Online) 13th April Morning Shift

Let $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$. Let $\mathrm{Re}(w) = 0$ be the circle C of radius 1 in the first quadrant touching the line $y = 1$ and the y-axis. If the curve $\mathrm{Im}(w) = 0$ intersects C at A and B, then $30(AB)^2$ is equal to _____

Q182. JEE Main 2023 (Online) 11th April Evening Shift

Let $S=\left\{z \in \mathbb{C}-\{i, 2 i\}: \frac{z^2+8 i z-15}{z^2-3 i z-2} \in \mathbb{R}\right\}$. If $\alpha-\frac{13}{11} i \in S, \alpha \in \mathbb{R}-\{0\}$, then $242 \alpha^2$ is equal to _____.

Q183. JEE Main 2023 (Online) 6th April Evening Shift

For $\alpha, \beta, z \in \mathbb{C}$ and $\lambda>1$, if $\sqrt{\lambda-1}$ is the radius of the circle $|z-\alpha|^2+|z-\beta|^2=2 \lambda$, then $|\alpha-\beta|$ is equal to _____.

Q184. JEE Main 2023 (Online) 30th January Morning Shift

Let $z=1+i$ and $z_1=\frac{1+i \bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$. Then $\frac{12}{\pi} \arg \left(z_1\right)$ is equal to _____.

Q185. JEE Main 2023 (Online) 29th January Evening Shift

Let $\alpha=8-14 i, A=\left\{z \in \mathbb{C}: \frac{\alpha z-\overline{\alpha z}}{z^2-(\bar{z})^2-112 i}=1\right\}$ and $B=\{z \in \mathbb{C}:|z+3 i|=4\}$. Then $\sum_{z \in A \cap B}(\operatorname{Re} z-\operatorname{Im} z)$ is equal to _____.

Q186. JEE Main 2022 (Online) 28th July Evening Shift

Let $z=a+i b, b \neq 0$ be complex numbers satisfying $z^2=\bar{z} \cdot 2^{1-|z|}$. Then the least value of $n \in N$, such that $z^n=(z+1)^n$, is equal to _____.

Q187. JEE Main 2022 (Online) 27th July Morning Shift

Let $S=\left\{z \in \mathbb{C}: z^2+\bar{z}=0\right\}$. Then $\sum_{z \in S}(\operatorname{Re}(z)+\operatorname{Im}(z))$ is equal to _____.

Q188. JEE Main 2022 (Online) 29th June Morning Shift

Let $S=\{z \in C:|z-2| \leq 1, z(1+i)+\overline{z}(1-i) \leq 2\}$. Let $|z-4 i|$ attain minimum and maximum values, respectively, at $z_1 \in S$ and $z_2 \in S$. If $5(|z_1|^2+|z_2|^2)=\alpha+\beta \sqrt{5}$, where α and β are integers, then the value of $\alpha+\beta$ is equal to _____.

Q189. JEE Main 2022 (Online) 28th June Evening Shift

Sum of squares of modulus of all the complex numbers z satisfying $\overline{z}=i z^2+z^2-z$ is equal to _____.

Q190. JEE Main 2022 (Online) 28th June Morning Shift

The number of elements in the set {z = a + ib $\in$ C : a, b $\in$ Z and 1 < | z $-$ 3 + 2i | < 4} is _____.

Q191. JEE Main 2022 (Online) 26th June Evening Shift

If $z^2 + z + 1 = 0, z \in C$, then

$\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ is equal to _____.

Q192. JEE Main 2022 (Online) 24th June Evening Shift

Let S = {z $\in$ C : |z $-$ 3| $\leq$ 1 and z(4 + 3i) + $\overline{z}$(4 $-$ 3i) $\leq$ 24}. If α + iβ is the point in S which is closest to 4i, then 25(α + β) is equal to _____.

Q193. JEE Main 2021 (Online) 1st September Evening Shift

If for the complex numbers z satisfying | z $-$ 2 $-$ 2i | $\leq$ 1, the maximum value of | 3iz + 6 | is attained at a + ib, then a + b is equal to _____.

Q194. JEE Main 2021 (Online) 31st August Morning Shift

A point z moves in the complex plane such that $\arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$, then the minimum value of $\left|z - 9\sqrt{2} - 2i\right|^2$ is equal to _____.

Q195. JEE Main 2021 (Online) 27th August Evening Shift

Let z_1 and z_2 be two complex numbers such that $\arg(z_1 - z_2) = \frac{\pi}{4}$ and z_1, z_2 satisfy the equation | z $-$ 3 | = Re(z). Then the imaginary part of $z_1 + z_2$ is equal to _____.

Q196. JEE Main 2021 (Online) 26th August Evening Shift

The least positive integer n such that $\frac{(2i)^n}{(1-i)^{n-2}}, i = \sqrt{-1}$ is a positive integer, is _____.

Q197. JEE Main 2021 (Online) 26th August Morning Shift

Let $z = \frac{1-i\sqrt{3}}{2}, i = \sqrt{-1}$. Then the value of $21 + \left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \ldots + \left(z^{21} + \frac{1}{z^{21}}\right)^3$ is _____.

Q198. JEE Main 2021 (Online) 27th July Evening Shift

If the real part of the complex number $z = \frac{3+2i\cos\theta}{1-3i\cos\theta}, \theta \in \left(0, \frac{\pi}{2}\right)$ is zero, then the value of $\sin^2 3\theta + \cos^2\theta$ is equal to _____.

Q199. JEE Main 2021 (Online) 25th July Evening Shift

The equation of a circle is $\text{Re}(z^2) + 2(\text{Im}(z))^2 + 2\text{Re}(z) = 0$, where $z = x + iy$. A line which passes through the center of the given circle and the vertex of the parabola, $x^2 - 6x - y + 13 = 0$, has y-intercept equal to _____.

Q200. JEE Main 2021 (Online) 18th March Morning Shift

Let z_1, z_2 be the roots of the equation $z^2 + az + 12 = 0$ and z_1, z_2 form an equilateral triangle with origin. Then, the value of $|a|$ is :

Q201. JEE Main 2021 (Online) 16th March Morning Shift

Let z and ω be two complex numbers such that $\omega = z\overline{z} - 2z + 2$, $\left|\frac{z+i}{z-3i}\right| = 1$ and $\text{Re}(\omega)$ has minimum value. Then, the minimum value of $n \in N$ for which ω^n is real, is equal to _____.

Q202. JEE Main 2021 (Online) 26th February Evening Shift

Let z be those complex numbers which satisfy

$|z + 5| \leq 4$ and $z(1 + i) + \overline{z}(1 - i) \geq -10$, $i = \sqrt{-1}$.

If the maximum value of $|z + 1|^2$ is $\alpha + \beta\sqrt{2}$, then the value of $(\alpha + \beta)$ is _____.

Q203. JEE Main 2021 (Online) 24th February Evening Shift

Let $i = \sqrt{-1}$. If $\frac{\left(-1+i\sqrt{3}\right)^{21}}{(1-i)^{24}} + \frac{\left(1+i\sqrt{3}\right)^{21}}{(1+i)^{24}} = k$, and $n = [|k|]$ be the greatest integral part of $|k|$. Then $\sum_{j=0}^{n+5}(j+5)^2 - \sum_{j=0}^{n+5}(j+5)$ is equal to _____.

Q204. JEE Main 2021 (Online) 24th February Morning Shift

If the least and the largest real values of α, for which the equation $z + \alpha|z - 1| + 2i = 0$ ($z \in C$ and $i = \sqrt{-1}$) has a solution, are p and q respectively; then $4(p^2 + q^2)$ is equal to _____.

Q205. JEE Main 2020 (Online) 3rd September Morning Slot

If $\left(\frac{1+i}{1-i}\right)^{\frac{m}{2}} = \left(\frac{1+i}{i-1}\right)^{\frac{n}{3}} = 1$, (m, n $\in$ N) then the greatest common divisor of the least values of m and n is _____ .

Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **A**	3. **A**	4. **D**	5. **B**
6. **D**	7. **A**	8. **C**	9. **D**	10. **A**
11. **D**	12. **C**	13. **D**	14. **D**	15. **D**
16. **A**	17. **B**	18. **A**	19. **D**	20. **C**
21. **A**	22. **A**	23. **B**	24. **A**	25. **C**
26. **A**	27. **D**	28. **A**	29. **A**	30. **B**
31. **C**	32. **B**	33. **B**	34. **B**	35. **A**
36. **A**	37. **C**	38. **C**	39. **B**	40. **B**
41. **A**	42. **C**	43. **D**	44. **D**	45. **B**
46. **D**	47. **B**	48. **C**	49. **A**	50. **D**
51. **B**	52. **B**	53. **A**	54. **C**	55. **B**
56. **C**	57. **D**	58. **D**	59. **D**	60. **C**
61. **A**	62. **D**	63. **A**	64. **D**	65. **A**
66. **D**	67. **B**	68. **B**	69. **D**	70. **B**
71. **A**	72. **C**	73. **C**	74. **A**	75. **C**
76. **D**	77. **C**	78. **D**	79. **D**	80. **C**
81. **A**	82. **C**	83. **A**	84. **B**	85. **C**
86. **B**	87. **D**	88. **D**	89. **D**	90. **A**
91. **B**	92. **D**	93. **A**	94. **B**	95. **B**
96. **D**	97. **B**	98. **C**	99. **B**	100. **B**
101. **D**	102. **A**	103. **A**	104. **C**	105. **A**
106. **A**	107. **D**	108. **B**	109. **A**	110. **D**
111. **A**	112. **B**	113. **B**	114. **D**	115. **D**
116. **D**	117. **C**	118. **C**	119. **D**	120. **D**
121. **B**	122. **D**	123. **C**	124. **B**	125. **C**
126. **A**	127. **C**	128. **B**	129. **D**	130. **C**
131. **C**	132. **A**	133. **A**	134. **D**	135. **B**
136. **D**	137. **A**	138. **B**	139. **C**	140. **A**
141. **D**	142. **B**	143. **B**	144. **A**	145. **B**
146. **C**	147. **A**	148. **C**	149. **A**	150. **A**

151. **A**	152. **C**	153. **A**	154. **D**	155. **D**
156. **C**	157. **C**	158. **C**	159. **C**	160. **A**
161. **B**	162. **A**	163. **D**	164. **B**	165. **B**
166. **C**	167. **B**			

Numerical

168. **5**	169. **49**	170. **11**	171. **22**	172. **10**
173. **31**	174. **9**	175. **36**	176. **3**	177. **49**
178. **13**	179. **20**	180. **5**	181. **24**	182. **1680**
183. **2**	184. **9**	185. **14**	186. **6**	187. **0**
188. **26**	189. **2**	190. **40**	191. **2**	192. **80**
193. **5**	194. **98**	195. **6**	196. **6**	197. **13**
198. **1**	199. **1**	200. **6**	201. **4**	202. **48**
203. **310**	204. **10**	205. **4**		